

A 40nm 2Mb ReRAM Macro with 85% Reduction in FORMING Time and 99% Reduction in Page-Write Time Using Auto-FORMING and Auto-Write Schemes

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Abstract

This work proposes (1) an auto-forming (AF) scheme to shorten the macro forming time (T_{FM-M}) and testing costs; (2) an auto-RESET (ARST) scheme to shorten page-RESET time ($T_{W-PAGE-RST}$) for expanding the applications of hidden-RESET operation in standby mode, and (3) an auto-SET (ASET) scheme to shorten page-write time (T_{W-PAGE}) combined with hidden-RESET scheme. A fabricated 40nm 2Mb ReRAM macro achieved 85+% reduction in T_{FM-M} , and 99+% reduction in T_{W-PAGE} for a page. For the first time, AF, ARST, and ASET schemes are demonstrated in silicon for ReRAM.

Keywords: ReRAM, forming, page-write

Introduction

ReRAM is a promising candidate for embedded nonvolatile memory (eNVM) for various cost-effective applications. However, as shown in Fig. 1, the design of ReRAM macros faces a number of challenges: (a) high testing costs and long device-stress times (T_{STRS}) due to long macro-level forming time (T_{FM-M}) to cover a wide-distribution in cell forming times (T_{FM}); and (b) long page-mode write-time (T_{W-PAGE}) and high write energy to store/refresh large-volume data due to wide distribution in SET/RESET times (T_{SET}/T_{RST}) [1]-[4] and the asymmetric relationship between T_{SET} and T_{RST} .

To overcome the challenges mentioned above, this work (Fig. 2) proposed (1) an AF scheme to reduce $T_{FM-MACRO}$, (2) an ARST scheme to shorten page-RESET time ($T_{W-PAGE-RST}$) and enable hiding $T_{W-PAGE-RST}$ in idle mode for wider applications, and (3) an ASET scheme to shorten page-write time (T_{W-PAGE}) combined with hidden-RESET scheme.

Proposed Auto Forming (AF) Scheme

Fig. 3 presents the concept of AF scheme, including two groups of column-multiplexor (YMUX) decoders (YMUXD), an auto-write controller (AWC), row-controller (RWC), and Triple-mode write-termination circuits (TM-WT). Each IO has two TM-WT modules (TM-WT0 and TM-WT1) when the column-multiplexor (YMUX) is large (the general case for NVM). The TM-WT expands previous dual-mode [1] or SET-only [2]-[4] write-termination circuits to three modes: FORMING, SET and RESET. In the main controller, there are two sets of AWCs (AWC0 and AWC1) and YMUXDs (YMUXD0 and YMUXD1), each of which is shared by all even/odd groups. Each AWC includes an address counter (AC), maximum delay controller (MDC), write-end-flag receiver (WENDFR), and auto-write control logics (AWCL). Fig. 4 shows the flow chart of AF scheme. At the beginning of an AF operation (AW_EN=1) on an even group (G0, column[63:0]), the AWC0 sends initial signals (WT_SEL_iniG0) to all TM-WT0s in G0 groups of all IOs. Each TM-WT0 then provides the required V_{FROM} to YMUXD0-selected SLs (i.e. column[0]).

When a ReRAM cell completes the FORMING operation, its TM-WT0 sends WENDFG0=1 to WENDFR0. When all WENDFG0[k-1:0] of all IOs are "1", then WENDFR0 sends a G0_DONE signal to AWCL0 and AC0. AC0 updates the address of YMUXD to change the column-address to the next column (i.e. column [1]). AWC0 then repeats the AF operation for the new column-address. Once all of the G0 columns in the same row have finished FORMING, AC0 sends ROW_G0=1 to RWC. When ROW_G0=ROW_G1=1, the RWC switches to the next row-address and begins AF on the next row (Fig. 5).

Note that when a cell fails to finish FORMING within a given time (T_{LIMIT}), the MDC forces a halt to the FORMING operation and outputs this fail-bit address. A similar operation is applied to the odd groups (AWC1 and G1, column [127:64]).

Auto-SET (ASET) and Auto-RESET (ARST) Schemes

Fig. 6 shows the concept of ARST and ASET. T_{RST} is usually much longer than T_{SET} and dominates the T_{W-PAGE} . Some system-level approaches for Flash memory hides long page-erase operation in system idle period. ReRAM could shorten T_{W-PAGE} by hiding page-RESET in the idle period and applying SET operations to selected cells within a page, as hidden-RESET page-write (HRPW). However, the long $T_{W-PAGE-RST}$ limited the applications of HRPW because not many systems have sufficient long idle periods to tolerate $T_{W-PAGE-RST}$.

Applying V_{RST} to V_{WRITE} of AF with bipolar switching of circuits in RESET path, the AF circuit can be reused as ARST to write all cells in a page from low-resistance state (LRS) to high-resistance state (HRS). For a page with p -cells, ARST reduces the $T_{W-PAGE-RST}$ from $pxT_{RST-MAX}$ to $pxT_{RST-AVG}$, where $T_{RST-MAX}/T_{RST-AVG}$ is the max./average T_{RST} of ReRAM cells.

To implement ASET scheme, this work add two address-skippers (ADSK) and an extra DIN-latch (DINL) to the AF circuitry. In ASET operations, the two DINLs store the data-in (DIN) bits for G0 and G1. When DINL=1, the ADSK outputs SKIP=1 to initiate a forced-termination in its group (G0/G1) and output WENDFG=1. When DIN=0 (no skip), the accessed column proceeds to the SET operation. After G0_DONE=1, the YMUX-address for G0 increases and requests a new DIN for G0 of all IOs. Fig. 7 shows the waveform of ASET.

Performances and Measurement Results

This work fabricated a 40nm 2Mb ReRAM macro with foundry's bipolar ReRAM devices (Fig. 8). The AF/ASET/ARST schemes consume less than 4% of the area overhead in a 2Mb macro. The captured waveforms (Fig. 9) showed that AF switches column addresses in response to variations in T_{FM} across IO-groups (G0/G1). AF reduced the T_{FOM-M} of a page by 85+% (Fig. 10a). ARST reduced the $T_{W-PAGE-RST}$ by 56.5% (Fig. 10b). ASET reduced the T_{W-PAGE} by 99+% (Fig. 10c) compared to that without HRPW scheme.

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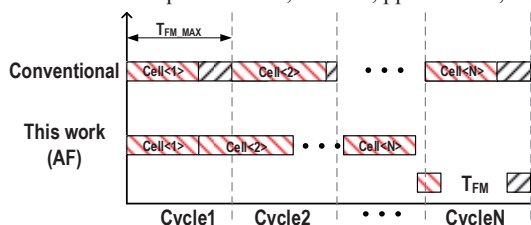


Fig. 2 Concept of Auto-Forming Scheme. (T_{FM} : Forming time per cell; T_{STRS} : Stress time per cell; T_{FM_MAX} : worst-case T_{FM})

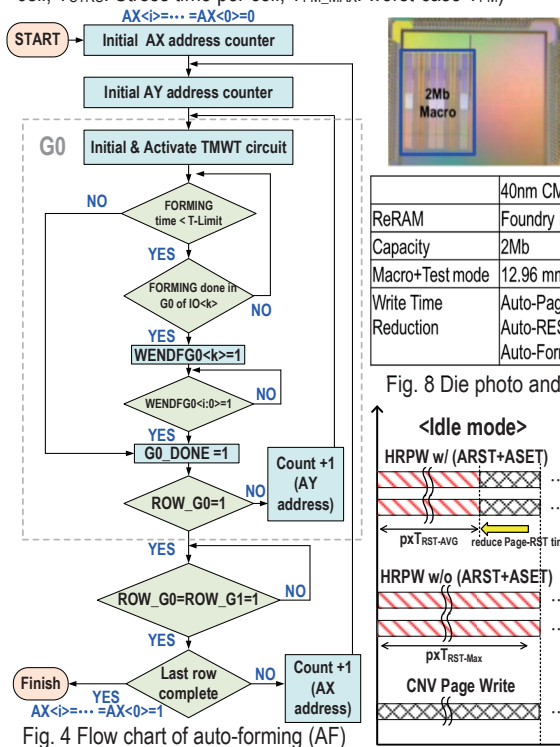


Fig. 4 Flow chart of auto-forming (AF)

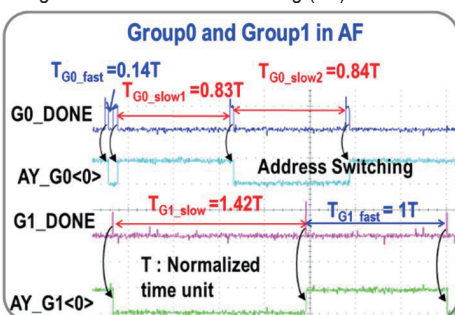


Fig. 9 Captured waveform of the self-switching on column-MUX address (AY) in AF operation

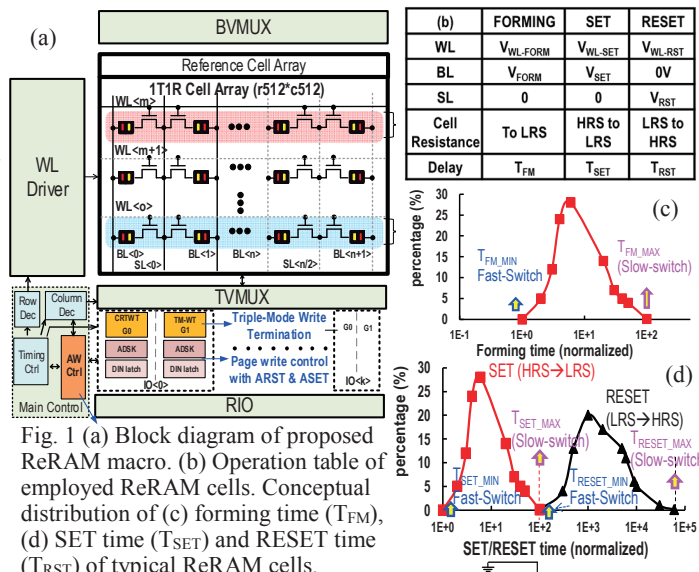


Fig. 1 (a) Block diagram of proposed ReRAM macro. (b) Operation table of employed ReRAM cells. Conceptual distribution of (c) forming time (T_{FM}), (d) SET time (T_{SET}) and RESET time (T_{RST}) of typical ReRAM cells.

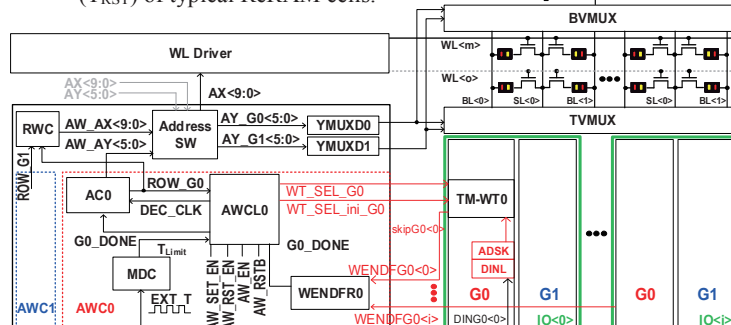


Fig. 3 Block Diagram of Auto-forming (AF), auto-SET (ASET), and auto-RESET (ARST)

ReRAM	40nm CMOS Logic Process
Capacity	Foundry ReRAM (Bipolar)
Macro+Test mode	2Mb
Write Time Reduction	Macro+Test mode: 12.96 mm ²
	Auto-Page-Write : 99.2%
	Auto-RESET : 56.51%
	Auto-Forming : 85.95%

Fig. 8 Die photo and chip summary

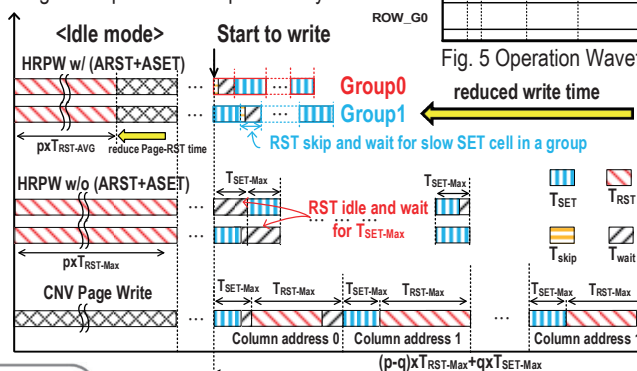


Fig. 6 Concept of ARST and ASET vs. conventional (CNV) page-write

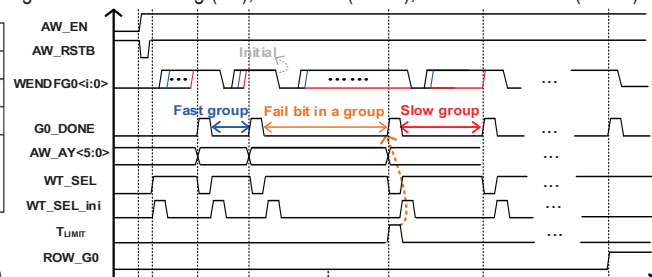


Fig. 5 Operation Waveform of Auto-Forming scheme

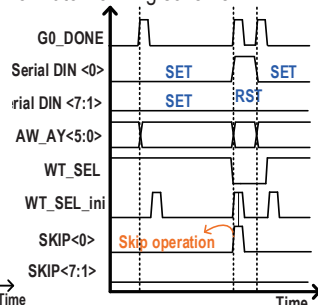


Fig. 7 Waveform of ASET and ARST

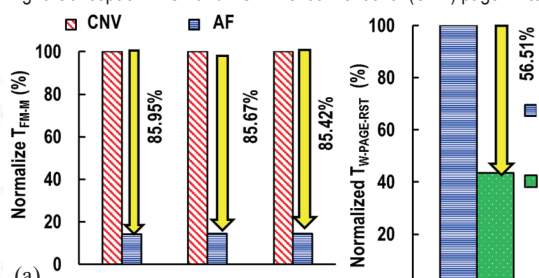


Fig. 10 Measured delay of (a) forming (T_{FM-M}), (b) page-RESET ($T_{W-PAGE-RST}$), and (c) page-write (T_{W-PAGE})